INCH-POUND

A-A-54850
16 April 1993

COMMERCIAL ITEM DESCRIPTION

LIGHT, SURGICAL, CEILING

Ventilated Dome, Rotation Type

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a ceiling mounted, operating and examining light.

Salient characteristics:

The surgical light shall be the ceiling-mounted type, and shall be suitable for use on all medical and dental facilities of all naval vessels, as well as other locations where low ceiling heights are found.

The light shall provide high intensity surgical quality light to permit illumination over examining, surgical, and delivery tables.

The light system shall consist of at least the following: a ceiling mounting tripod with leveling capabilities, suspension tube, yoke assembly, light head with sterilizable handle, lamp bulb, heat dissipating system, color correction system, controllable lamp positioning system, position locking device, on/off switch, and electrical plug which shall combine to form one item.

The center line of the lamphead shall be approximately 13-1/2 inches from the center line of the vertical suspension tube. The horizontal extension bar shall be approximately 6-1/2 inches long and allow mounting of the yoke in a plane 45 degrees below horizontal. The center line of the suspension tube to outer edge of lamphead shall be approximately 20-1/2 inches.

AMSC N/A FSC 6530

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The minimum distance from ceiling to the face of the lamphead shall be 16 inches.

Lamphead enclosure shall be of polycarbonate or equal material.

The lamphead shall be approximately 13 inches in diameter. Thickness of lamphead from lens cover to top of ventilated dome shall not exceed 7 inches.

The lamphead shall contain a single 10.5 inch diameter ultem polytherimide molded reflector with a color correcting infrared transmitting coating vacuum deposited on its surface.

Each lamphead shall contain a 120 volt, 150 watt halogen lamp with an average rated life of 1000 hours.

The finish of the light system shall be highly fungus, mildew, corrosion and salt-spray resistant. The finish shall meet the following ASTM Standards: D3359 Adhesion, B499 Dry Mill Thickage, D523 Glass.

Unit shall have the highest quality, commercially available, easily cleaned, washable finish which incorporates at least the following paint processes and finish for:

a. Tripod, suspension tube, yoke assembly, and arms:

Iron phosphate wash.
Alkaline spray clean.
Rinse.
Zinc Phosphate.
Rinse.
Dry.
Non-chromic acid rinse.
Polyurethane color coat.
Dry.
Polyurethane texture coat.
Bake.

b Lamphead

Polyurethane color coat.
Air dry.
Polyurethane texture coat.
Bake.

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Horizontal rotation of lamp about the vertical suspension tube shall be 310 degrees. Yoke rotation about the horizontal suspension tube shall be 180 degrees. Lamphead rotation about its own axis shall be 180 degrees. Lamphead rotation range about its mounting fork shall be 45 degrees in each direction above and below horizontal.

The arm, yoke, and lamphead of the light system shall be counterbalanced throughout its full travel; lamphead position shall have capability of being maintained without the use of locking devices. However, a method of locking the lamp position shall be provided for those times when it is desirable to do so.

Positioning of the lamphead and arm assembly shall be controllable in two ways: by a detachable sterilizable control handle, and by a non-sterilizable handle on the lamphead.

Light intensity from the lamphead shall be approximately 4000 footcandles at the center spot, plus or minus 10 percent, 36 inches from the lower rim of the lens cover with the lamphead in a horizontal position.

Illumination shall be distributed over a field of at least 144 square inches.

The light shall have universal prefocus design so that the beam will be in proper focus through its normal range of use, 36 inches. Need for adjustment of light focusing shall be minimal.

Light beam shall be color corrected to provide a correlated color temperature of approximately 4800 degrees Kelvin, plus or minus 10 percent, suitable for Kodachrome Type A film without additional filters. The irradiance factor shall not exceed 5.0 microwatts per square centimeter per footcandle at a distance of 36 inches from the lens.

Color correction shall be achieved by the use of a combination of filters: cylindrical borosilicate glass filters, and dichroic coating filter applied to the ultem fluted reflector.

Dichroic filters shall be blue-green color correcting borosilicate glass coatings.

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The ultem fluted reflectors shall provide diffusion of light to keep glare at a minimum.

Dissipation of heat shall be achieved by absorption into the glass filter, by transmission through the dichroic coating on the reflector, and by convection through dome assembly.

The lens cover shall diffuse the light from the light source.

Light system shall be designed to permit installation by a single individual in 10 minutes or less.

The light shall be field repairable with common hand tools, with lamp replacement taking less than one minute and requiring no tools.

The light shall be supplied with an on/off switch at the lamphead.

The electrical characteristics shall be 120 volts and 50/60 Hz.

The maximum weight allowable shall be 18 pounds.

The light system shall meet the following Standards and/or have the following approvals:

* Nationals Electric Code (National Fire Codes, Vol. V) as prescribed in National Fire Protection Association pamphlet NFPA No 56A

* Underwriters Laboratory Code 544 or meet the standards of any other Government approved testing laboratory, and be so labeled.

Installed unit shall conform to standards established for Grade A shock testing for shipboard use.

Workmanship. The light system shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One surgical ceiling light system, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one light system.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one light system.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

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* paint finish cross hatch scratch test
* fungus, mildew, and corrosion resistance
* light intensity
* illumination field size
* color temperature
* irradiance factor
* shock test

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6530-01-321-5587

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6530-2389

Location: LIGHT1DC.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2 The submitter of this form must complete blocks 4, 5, 6, and 7.

3 The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54850	2. DOCUMENT DATE (YYMMDD): 16 April 1993

3. DOCUMENT TITLE: LIGHT, SURGICAL, CEILING Ventilated Dome, Rotation Type

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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